THE EXACT ACTIVITY THRESHOLD FOR SPECTRAL INDEPENDENCE OF THE HARD-CORE MODEL ON TREES

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Abstract. For the hard-core model on finite forests with vertex-dependent activities we determine the exact activity threshold for uniform spectral independence and for an optimal spectral gap of the Glauber dynamics. Let $L = 5^{-1/2}(1-5^{-1/2})^{-6} \approx 15.6737$. If all activities are at most a constant $\Lambda < L$, then the correlation matrix of the occupation indicators has operator norm bounded in terms of Λ alone. The bound persists under every pinning and every decrease of the activities, so the model is completely spectrally independent in the sense of Chen, Feng, Yin and Zhang, and the single-site heat-bath dynamics has spectral gap at least c_Λ/n. No assumption on the degrees is made. Conversely, for every constant activity $\lambda \geq L$ there are finite trees with free boundary conditions on which the correlation matrix has arbitrarily large norm and n times the spectral gap is arbitrarily small. This closes the gap between the thresholds e^2 and 28.15 obtained by Chen, Yang, Yin and Zhang. The main new ingredient is an inequality for a cubic potential that holds for an arbitrary number of children. A convexity argument reduces it to finitely many one-variable estimates, which we verify by interval arithmetic with rational endpoints.

1. Introduction

Let $F = (V, E)$ be a finite forest and let $\lambda = (\lambda_v)_{v \in V}$ be positive numbers, the *activities*. The hard-core model on F is the probability measure

$$\mu_{F,\lambda}(I) = \frac{1}{Z_{F,\lambda}} \prod_{v \in I} \lambda_v, \qquad I \subseteq V \text{ independent}, \tag{1.1}$$

where $Z_{F,\lambda}$ is the normalizing constant. The *Glauber dynamics* (heat-bath dynamics) chooses a vertex uniformly at random and resamples its occupation from the conditional law given all other vertices. We write $\mathrm{gap}(\mu)$ for the spectral gap of this chain and $n = |V|$.

For graphs of maximum degree $\Delta \geq 3$, spectral independence, introduced by Anari, Liu and Oveis Gharan [1], gives rapid mixing of the Glauber dynamics for the hard-core model with activities below the uniqueness threshold $\lambda_c(\Delta) = (\Delta-1)^{\Delta-1}/(\Delta-2)^{\Delta}$; see [1, 3, 5] and the references there. On trees the situation is different, since sampling is tractable for every activity, and the relevant question is for which activities the Glauber dynamics has the optimal spectral gap $\Omega(1/n)$ on all trees, without any restriction on the degree. For the relaxation time of the Glauber dynamics on trees of bounded degree see Berger, Kenyon, Mossel and

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Peres [2]. Restrepo, Štefankovič, Vera, Vigoda and Yang [9] constructed trees on which the dynamics slows down polynomially when the activity is large; the constant implicit in their construction is about 28 (see [6, Section 1]). Efthymiou, Hayes, Štefankovič and Vigoda [6] proved approximate tensorization of variance on all trees for $\lambda \le 1.1$, and, combining spectral independence with the boosting theorem of Chen, Feng, Yin and Zhang [3, Theorem 1.9], an optimal relaxation time for $\lambda \le 1.3$. Chen, Yang, Yin and Zhang [4] introduced an approximate inverse of the influence matrix and proved spectral independence and an optimal spectral gap on all trees for $\lambda < e^2 \approx 7.389$ [4, Theorem 7]. In the other direction they showed that spectral independence is unbounded on trees for $\lambda > 28.15$ [4, Theorem 8].

In this paper we close the gap between these two thresholds. Put

$$q_* = \frac{1}{\sqrt{5}}, \qquad L = \frac{q_*}{(1-q_*)^6} = 15.6736764833\ldots. \tag{1.2}$$

We use the notion of complete spectral independence of [3], which is based on the absolute influence matrix and requires a bound after every pinning and every decrease of the activities; it is recalled in Definition 2.5.

Theorem 1.1. (a) *Let $0 < \Lambda < L$. There are constants $\eta_\Lambda < \infty$ and $c_\Lambda > 0$, depending only on Λ, such that the following holds for every finite forest F on $n \ge 1$ vertices and all activities $0 < \lambda_v \le \Lambda$. The hard-core measure $\mu_{F,\lambda}$ is completely η_Λ-spectrally independent, and*

$$\operatorname{gap}(\mu_{F,\lambda}) \;\ge\; \frac{c_\Lambda}{n}.$$

(b) *Let $\lambda \ge L$ be constant. For every $M > 0$ there is a finite tree T on n vertices such that the hard-core measure μ on T with activity λ at every vertex satisfies $\rho(\Psi_\mu) \ge M$ and $n \cdot \operatorname{gap}(\mu) \le 1/M$.*

Here Ψ_μ is the influence matrix of the unpinned measure μ, and ρ denotes the spectral radius. In part (b) no pinning and no external field are used: the constructed trees carry free boundary conditions. The activities in part (a) are arbitrary in $(0, \Lambda]$; in particular, there is no lower bound on them and no bound on the degrees.

Part (a) is a consequence of a bound on the correlation matrix. For a measure on $\{0,1\}^V$ without almost surely constant coordinates let X_v be the occupation indicator of v, $a_v = \mathbb{E}X_v$, $s_v = \operatorname{Var} X_v$, and

$$R_{uv} = \frac{\operatorname{Cov}(X_u, X_v)}{\sqrt{s_u s_v}} \qquad (u, v \in V).$$

Theorem 1.2. *Let $0 < \Lambda < L$. There is $\eta_\Lambda < \infty$ such that for every finite forest and all activities $0 < \lambda_v \le \Lambda$ the correlation matrix of the hard-core measure satisfies $\|R\|_2 \le \eta_\Lambda$.*

Since pinnings and decreases of activities of a hard-core measure on a forest produce hard-core measures on smaller forests with activities at most Λ, Theorem 1.2 is automatically uniform over them. On a forest the absolute influence matrix is similar to $|R| - I$, and $|R|$ has the same spectrum as R (Lemma 2.4). Theorem 1.2 therefore gives complete spectral independence, and the spectral gap follows from [3, Theorem 1.9] together with the variance tensorization of [6, Section 3] at small activities.

The cubic partition inequality. The proof of Theorem 1.2 rests on a scalar inequality. Fix the constants $c = 2/125$ and $z \approx 2.8310825$ defined in (3.1) below, and set

$$\Phi(y) = y\big(1 + c(y - z)^2\big), \qquad X = \log(1 + L). \tag{1.3}$$

For $0 < x \le X$ put $r = e^x - 1$, $q = r/(1 + r)$, and

$$f(x) = q^2\, \Phi\big(\log(L/r)\big), \qquad f(0) = 0. \tag{1.4}$$

In the hard-core recursion on a rooted tree, r is the occupation odds of a child in its own subtree, $x = \log(1 + r)$ is its contribution to the potential of its parent, and $\log(L/r)$ dominates the potential of the child itself.

Theorem 1.3. *For every $m \ge 1$ and all $x_1, \dots, x_m \in [0, X]$,*

$$\sum_{i=1}^{m} f(x_i) \;\le\; \Phi\Big(\sum_{i=1}^{m} x_i\Big).$$

The inequality is sharp: for five equal entries $x_i = x_* = -\log(1 - q_*)$ both sides coincide. Since m is arbitrary, the resulting bounds do not depend on the degrees. The proof in Section 3 shows by a convexity argument that a hypothetical violating list may be assumed to consist of equal positive entries, and that the number of entries must then be five. What remains are finitely many inequalities in one real variable (Proposition 3.2); these are verified in Section 7 by interval arithmetic with rational endpoints, with analytic arguments near the singular endpoint.

From Theorem 1.3 we derive in Section 3 a strict contraction with factor $\gamma(\Lambda) < 1$ for every activity cap $\Lambda < L$ (Lemma 3.7). In Section 4 it is applied, generation by generation, to the sums $\sum Q^2 \Phi(y)$, where Q is a product of conditional occupation probabilities along a path and y is the potential at the end of the path. This yields geometric decay of the squared correlations between a vertex and its descendants at a given distance. A triangular factorization of the correlation matrix (Lemma 4.3) turns this decay into a bound on $\|R\|_2$.

The potential approach follows Fang, Lu, Nevo, Yao and Zheng [7], who used the potential $y = \sum \log(1 + r)$ together with the scalar inequality [7, Lemma 4.2] to control sums of powers of path products, in their study of the size of a random independent set at activities up to 12. The linear potential $\Phi(y) = y$ corresponds to the exponent two in [7, (4.5)]; for it the analogue of Theorem 1.3 reduces to a single-variable inequality, which numerically holds only for caps slightly below L (Remark 3.6). The cubic correction in (1.3) closes this gap.

The upper threshold. Part (b) of Theorem 1.1 is obtained by the mechanism of [4, Section 5.2]: a regular core tree is attached at its leaves to finite trees whose root marginals approximate a fixed point of the hard-core recursion. These trees exist by [9, Section 5] in the form of [4, Lemma 57]. We use a core in which the root has six children and every other internal vertex has five. The core law then converges to a stationary broadcast process in which a vertex is occupied with probability p when its parent is unoccupied, where $p = \lambda(1 - p)^6$, and the Rayleigh quotient of the inverse correlation matrix at the test vector of [4, (21)] tends to zero as the height of the core grows, as soon as $5p^2 \ge 1$, that is, as soon as $\lambda \ge L$. With branching number b in place of five the same construction requires $bp^2 \ge 1$ with $p = \lambda(1 - p)^{b+1}$, that is, $\lambda \ge \lambda_b := b^{-1/2}(1 - b^{-1/2})^{-(b+1)}$. One has $\lambda_2 \approx 28.14$, which underlies the bound 28.15 in [4, Theorem 8], $\lambda_3 \approx 18.09$, $\lambda_4 = 16$, $\lambda_5 = L$ and

$\lambda_6 \approx 16.07$. Since the construction applies for every integer $b \geq 2$ (Remark 6.6), Theorem 1.1(a) gives $\lambda_b \geq L$ for all such b. Thus, among the integers $b \geq 2$, λ_b is minimal at $b = 5$, where $\lambda_5 = L$.

Dependencies. Besides standard facts about the hard-core model and finite Markov chains, the proof uses three published results that we do not reprove: the existence of finite trees with prescribed approximate root marginal [4, Lemma 57] (from [9, Section 5]), the boosting theorem [3, Theorem 1.9], and the variance tensorization for trees with activities at most 1.1 from [6, Section 3]. The first enters only part (b) of Theorem 1.1, and the last two only its spectral gap assertion; Theorems 1.2 and 1.3 and the spectral independence assertion use none of them. Theorem 1.3 depends on finitely many interval computations, described in Section 7 and Appendix A. Theorem, equation and section numbers of [3, 4, 5, 6, 7] refer to the arXiv versions listed in the references.

Organization. Section 2 collects notation and standard facts about the hard-core model on forests. Section 3 proves Theorem 1.3, assuming the scalar estimates of Proposition 3.2, and deduces the strict contraction. Section 4 proves Theorem 1.2 and the spectral independence part of Theorem 1.1. Section 5 proves the spectral gap bound, and Section 6 proves Theorem 1.1(b). Section 7 proves Proposition 3.2.

2. Preliminaries

2.1. **Notation.** Throughout, $F = (V, E)$ is a finite forest with $n = |V| \geq 1$ and $\lambda = (\lambda_v)_{v \in V}$ are positive activities. We write $\mu = \mu_{F,\lambda}$ for the measure (1.1) and identify an independent set I with its indicator vector $(X_v)_{v \in V} \in \{0,1\}^V$. Since every activity is positive, both the empty set and every singleton have positive probability, so $0 < a_v < 1$ for every v, where $a_v = \mu(X_v = 1)$. We put $s_v = a_v(1-a_v)$, and let R be the correlation matrix defined in the introduction. Operator norms are Euclidean.

We choose a root in every component of F. For a vertex v we write T_v for the subtree of descendants of v (including v), $\mathcal{C}(v)$ for the set of children of v, and, if v is not a root, v^- for its parent. For a descendant u of v let $\operatorname{dist}(v,u)$ be the graph distance. All statements below hold for an arbitrary choice of roots.

For $v \in V$ let Z_v^1 and Z_v^0 be the total weights of the independent sets of T_v that contain v, respectively do not contain v, computed with the activities λ. Define

$$r_v = \frac{Z_v^1}{Z_v^0}, \qquad q_v = \frac{r_v}{1+r_v}, \qquad x_v = \log(1+r_v), \qquad y_v = \sum_{u \in \mathcal{C}(v)} x_u. \tag{2.1}$$

Thus r_v is the occupation odds and q_v the occupation probability of v under the hard-core measure on T_v alone. For a descendant u of v let

$$Q(v,u) = \prod_w q_w,$$

the product over the vertices w of the path from v to u other than v; in particular $Q(v,v) = 1$.

2.2. **Recursion, Markov property and marginals.**

Lemma 2.1. *Let $w \in V$. Conditionally on $X_w = 0$, the restrictions of X to the components of $F - w$ are independent, and each is distributed as the hard-core measure on that component with the activities λ. Conditionally on $X_w = 1$, every*

neighbour of w is unoccupied. In both cases, the restriction of X to a component C of $F - w$ is conditionally independent of the restriction of X to $V \setminus (C \cup \{w\})$.

Proof. An independent set I with $w \notin I$ is the disjoint union of independent sets of the components of $F - w$, and its weight is the product of their weights. An independent set containing w is $\{w\}$ together with an independent set of $F - w$ avoiding the neighbours of w, which is again a disjoint union over the components of $F - w$, each component C contributing an independent set of C avoiding the unique neighbour of w in C, if any. The weights factor in the same way. Both assertions follow. □

Lemma 2.2. (i) *For every v, $r_v = \lambda_v \prod_{u \in \mathcal{C}(v)} (1 + r_u)^{-1} = \lambda_v e^{-y_v}$. In particular $0 < r_v \le \lambda_v$ and $y_v = \log(\lambda_v / r_v) \ge 0$.*

(ii) *For a child u of w one has $\mu(X_u = 1 \mid X_w = 0) = q_u$ and $\mu(X_u = 1 \mid X_w = 1) = 0$.*

(iii) *If every activity is at most Λ, then for every v*

$$\frac{q_v}{1+\Lambda} \le a_v \le q_v, \qquad s_v \ge \frac{q_v}{(1+\Lambda)^2}, \qquad s_v \le \Lambda e^{-y_v}.$$

Proof. (i) An independent set of T_v not containing v is a disjoint union of independent sets of the trees T_u, $u \in \mathcal{C}(v)$, so $Z_v^0 = \prod_u (Z_u^0 + Z_u^1)$. An independent set containing v consists of v and independent sets of the T_u avoiding their roots, so $Z_v^1 = \lambda_v \prod_u Z_u^0$. Dividing gives the recursion, and $\prod_u (1 + r_u)^{-1} = e^{-y_v}$ by (2.1).

(ii) The tree T_u is a component of $F - w$, so the first identity follows from Lemma 2.1. The second is the hard-core constraint.

(iii) If v is a root, put $t = 0$. Otherwise delete the edge vv^-, let F' be the component of the resulting forest that contains v^-, and let t be the occupation odds of v^- under the hard-core measure on F'; by (i), applied to F' rooted at v^-, we have $0 < t \le \lambda_{v^-} \le \Lambda$. Summing the weights of independent sets of F according to the occupation of v and v^-, and dividing by the product of the weights of the empty configurations at v in T_v and at v^- in F', we obtain

$$a_v = \frac{r_v}{1 + r_v + t}.$$

Since $1 + r_v + t \le (1 + r_v)(1 + \Lambda)$, this gives $q_v/(1+\Lambda) \le a_v \le q_v$. Moreover $q_v \le r_v \le \Lambda$ implies $1 - a_v \ge 1 - q_v \ge (1+\Lambda)^{-1}$, so $s_v = a_v(1 - a_v) \ge q_v(1+\Lambda)^{-2}$. Finally $s_v \le a_v \le q_v \le r_v = \lambda_v e^{-y_v} \le \Lambda e^{-y_v}$. □

Lemma 2.3. *Let u be a descendant of v with $\operatorname{dist}(v, u) = d$. Then*

$$\mathbb{E}[X_u \mid X_v = 1] - \mathbb{E}[X_u \mid X_v = 0] = (-1)^d Q(v, u),$$

and consequently

$$\operatorname{Cov}(X_v, X_u) = (-1)^d s_v Q(v, u), \qquad R_{vu} = (-1)^d \sqrt{s_v / s_u}\, Q(v, u). \tag{2.2}$$

Proof. Let $v = w_0, w_1, \dots, w_d = u$ be the path from v to u, and put $g_k(\xi) = \mathbb{E}[X_{w_k} \mid X_v = \xi]$ for $\xi \in \{0, 1\}$. For $k \ge 1$ the tree T_{w_k} is a component of $F - w_{k-1}$ and $v \notin T_{w_k}$. By Lemma 2.1 and Lemma 2.2(ii),

$$\mathbb{E}[X_{w_k} \mid X_{w_{k-1}}, X_v] = q_{w_k}(1 - X_{w_{k-1}}),$$

hence $g_k(\xi) = q_{w_k}(1 - g_{k-1}(\xi))$ and $g_k(1) - g_k(0) = -q_{w_k}(g_{k-1}(1) - g_{k-1}(0))$. Since $g_0(1) - g_0(0) = 1$, the first identity follows. For a $\{0,1\}$-valued X_v and any bounded Y one has $\operatorname{Cov}(X_v, Y) = s_v(\mathbb{E}[Y \mid X_v = 1] - \mathbb{E}[Y \mid X_v = 0])$, which gives (2.2). □

Lemma 2.4. *Let $u \neq v$. If u and v lie in different components of F, then $R_{uv} = 0$. Otherwise $R_{uv} \neq 0$ and its sign is $(-1)^{\operatorname{dist}(u,v)}$. Consequently, if $J = \operatorname{diag}((-1)^{\operatorname{depth}(v)})$, where the depth is measured from the root of the component of v, then the entrywise absolute value of R is $|R| = JRJ$, and $|R|$ has the same eigenvalues as R.*

Proof. Different components are independent. If u and v lie in the same component, root that component at v; this does not change R, and Lemma 2.3 gives the sign, since $Q(v,u) > 0$. Finally $\operatorname{depth}(u) + \operatorname{depth}(v) \equiv \operatorname{dist}(u,v) \pmod 2$ for vertices of one component, so $(JRJ)_{uv} = |R_{uv}|$, and $J^{-1} = J$. □

2.3. **Spectral independence and the Glauber dynamics.** We recall the definitions of [3, Section 1.2], translated from $\{-1,+1\}$ to $\{0,1\}$ with $+1$ corresponding to an occupied vertex. Let ν be a probability measure on $\{0,1\}^V$ with support $\Omega(\nu)$. For $\Lambda' \subseteq V$ and a feasible partial configuration $\sigma \in \{0,1\}^{\Lambda'}$ let ν^σ be ν conditioned on $X_{\Lambda'} = \sigma$. For local fields $\phi \in (0,1]^V$ let $\nu^{(\phi)}$ be the measure with $\nu^{(\phi)}(\omega) \propto \nu(\omega) \prod_{v:\,\omega_v=1} \phi_v$.

Definition 2.5 ([3, Definitions 1.5 and 1.8])**.** The (absolute) influence matrix Ψ_{ν^σ} is indexed by $V \setminus \Lambda'$ and has zero diagonal. For distinct $u, v \in V \setminus \Lambda'$, $\Psi_{\nu^\sigma}(u,v) = 0$ if X_u is ν^σ-almost surely constant, and otherwise

$$\Psi_{\nu^\sigma}(u,v) = \big|\nu^\sigma(X_v = 1 \mid X_u = 1) - \nu^\sigma(X_v = 1 \mid X_u = 0)\big|.$$

The measure ν is *η-spectrally independent* if $\rho(\Psi_{\nu^\sigma}) \leq \eta$ for all Λ' and all feasible σ, and *completely η-spectrally independent* if $\nu^{(\phi)}$ is η-spectrally independent for every $\phi \in (0,1]^V$.

For two-point laws the total variation distance in [3, Definition 1.5] is the difference of the probabilities of one point, so this is the definition of [3]. When X_u and X_v are both nonconstant, $\Psi_{\nu^\sigma}(u,v) = |\operatorname{Cov}(X_u, X_v)| / \operatorname{Var}(X_u)$, the moments being taken under ν^σ.

The Glauber dynamics for ν picks $v \in V$ uniformly at random and replaces X_v by a sample from ν conditioned on $X_{V\setminus\{v\}}$ [3, Section 1]. It is reversible with respect to ν, and its transition matrix is positive semidefinite. Its spectral gap $\operatorname{gap}(\nu)$ is one minus the second largest eigenvalue, with $\operatorname{gap}(\nu) = 1$ if $|\Omega(\nu)| = 1$. By the variational characterization (see [8, Chapter 13]),

$$\operatorname{gap}(\nu) = \inf_g \frac{\mathcal{E}_\nu(g,g)}{\operatorname{Var}_\nu(g)}, \qquad \mathcal{E}_\nu(g,g) = \frac{1}{n}\sum_{v\in V} \mathbb{E}_\nu\big[\operatorname{Var}_\nu(g \mid X_{V\setminus\{v\}})\big], \tag{2.3}$$

the infimum being over functions g on $\Omega(\nu)$ that are not constant. Pinned measures ν^σ are regarded as measures on $\{0,1\}^V$, so that the Glauber dynamics for ν^σ still selects among all n vertices; this is the convention of [3, (4)].

Lemma 2.6. *Let $\phi \in (0,1]^V$, $\Lambda' \subseteq V$, and let $\sigma \in \{0,1\}^{\Lambda'}$ be feasible for $\nu = \mu^{(\phi)}_{F,\lambda}$. Let U be the set of vertices of $V \setminus \Lambda'$ that have no neighbour $w \in \Lambda'$ with $\sigma_w = 1$. Under ν^σ the coordinates in $V \setminus (\Lambda' \cup U)$ are almost surely 0, and the restriction of X to U is distributed as the hard-core measure on the induced forest $F[U]$ with activities $(\phi_v \lambda_v)_{v\in U}$.*

Proof. The measure ν is the hard-core measure on F with activities $\phi_v\lambda_v$. Its configurations agreeing with σ are exactly the sets $\{w \in \Lambda' : \sigma_w = 1\} \cup J$ with

J an independent set of $F[U]$, and the weight of such a set is a constant times $\prod_{v\in J}\phi_v\lambda_v$. □

3. The cubic partition inequality

3.1. **Constants and the potential.** Let q_* and L be as in (1.2), and define

$$\begin{aligned} c &= \frac{2}{125}, \qquad x_* = -\log(1-q_*), \qquad y_* = 5x_*, \\ d_* &= \frac{2q_*(1-q_*)}{1+q_*}, \qquad D = d_* - \frac{1}{y_*}, \qquad t = \frac{D}{c\bigl(1+\sqrt{1-D^2/c}\bigr)}, \\ z &= y_* - t, \qquad X = \log(1+L), \qquad A = \log\tfrac{5}{2}. \end{aligned} \tag{3.1}$$

Proposition 3.2(P1) below shows that $D>0$ and $D^2<c$, so that $t>0$ is well defined. Numerically,

$$\begin{aligned} x_* &= 0.5927836\ldots, \qquad y_* = 2.9639180\ldots, \qquad d_* = 0.3416407\ldots, \\ D &= 0.0042495\ldots, \qquad t = 0.1328355\ldots, \qquad z = 2.8310824\ldots. \end{aligned}$$

Let Φ be the cubic (1.3). Since $5q_*^2=1$, we also have $d_* = 2(1-q_*)/(5q_*+1)$.

Lemma 3.1. *$2ct/(1+ct^2) = D$.*

Proof. Put $s=\sqrt{1-D^2/c}\in(0,1)$, so that $t = D/(c(1+s))$. Then $ct^2 = D^2/(c(1+s)^2) = (1-s^2)/(1+s)^2 = (1-s)/(1+s)$, hence $1+ct^2 = 2/(1+s)$, while $2ct = 2D/(1+s)$. □

For $0<x\le X$ we use the variables

$$\begin{aligned} r &= e^x - 1 \in (0,L], \qquad q = \frac{r}{1+r} = 1-e^{-x}, \\ y &= \log\frac{L}{r} = \log L + \log(1-q) - \log q \ \ge 0, \end{aligned} \tag{3.2}$$

and define $f(x) = q^2\Phi(y)$ as in (1.4), with $f(0)=0$. The point $x=A$ corresponds to $q=3/5$, and $x=x_*$ to $q=q_*$. Note that $y(x_*) = \log\bigl(L(1-q_*)/q_*\bigr) = -5\log(1-q_*) = y_*$.

3.2. **The scalar estimates.** The following finitely many one-variable statements are proved in Section 7. Primes denote derivatives with respect to x.

Proposition 3.2. (P1) *$2.8310 < z < 2.8311$, $D>0$, $D^2<c$, $1/2<x_*<31/50$, $L/(1+L)<941/1000$, and $9/10<A<X$.*
(P2) *$f(x) < \frac{1001}{1000}\,x$ for $0<x\le X$.*
(P3) *$f(x) < \frac{99}{100}\,x$ for $A\le x\le X$.*
(P4) *$f(x)<x$ for $0<x\le 13/25$ and for $5/8\le x\le X$.*
(P5) *$f''''(x)>0$ for $0<x\le A$, and $f''(A)<0$.*
(P6) *The function $G(x)=\Phi(5x)-5f(x)$ satisfies $G''(x)>0$ for $1/2\le x\le 31/50$.*

Lemma 3.3. (i) *$\Phi(0)=0$ and $\Phi(y)\ge y$ for $y\ge 0$.*
(ii) *$\Phi'(y) = 1+c(3y^2-4zy+z^2) \ge 1-cz^2/3 > 0$ for all real y; in particular Φ is strictly increasing.*
(iii) *$\Phi(Y)\ge\frac{1001}{1000}\,Y$ for every $Y\ge 0$ with $Y\notin(5/2,31/10)$.*
(iv) *$\Phi'(y_*)/\Phi(y_*) = d_*$.*

Proof. (i) is clear. (ii) The quadratic $3y^2 - 4zy + z^2$ has minimum $-z^2/3$, and $cz^2/3 < 3c < 1$ by (P1). (iii) For $Y \geq 0$ we have $\Phi(Y)/Y = 1 + c(Y-z)^2$ if $Y > 0$. By (P1), if $Y \leq 5/2$ or $Y \geq 31/10$ then $|Y - z| > 1/4$, and $c/16 = 1/1000$. The case $Y = 0$ is trivial. (iv) We have $\Phi'(y)/\Phi(y) = 1/y + 2c(y-z)/(1 + c(y-z)^2)$ for $y > 0$. At $y = y_*$ we have $y - z = t$, and Lemma 3.1 gives $\Phi'(y_*)/\Phi(y_*) = 1/y_* + D = d_*$. □

Lemma 3.4. *If $0 < q \leq 1/100$, then with $u = \log(L/q)$,*

$$\frac{f(x)}{x} \leq q\,(u + cu^3) \leq \frac{138}{1000}.$$

Moreover $f(x)/x \to 0$ as $x \to 0$. In particular f is continuous on $[0, X]$, and it is infinitely differentiable on $(0, X]$.

Proof. We have $x = -\log(1-q) \geq q$ and $y = \log(L(1-q)/q) \leq u$. Since $L > 15$, $u \geq \log(1500) > 7 > z$, so $0 < u - z < u$ and $\Phi(u) \leq u + cu^3$. As Φ is increasing, $f(x)/x \leq q^2\Phi(u)/q \leq q(u + cu^3)$. As a function of q, with $du/dq = -1/q$,

$$\frac{d}{dq}\, q(u + cu^3) = (u-1) + cu^2(u-3) > 0 \qquad (u \geq 3),$$

so the bound is increasing on $(0, 1/100]$. Since $q\log^3(1/q) \to 0$ as $q \to 0$, the bound tends to 0. At $q = 1/100$ we have $u = \log(100L) < \log 1568 < 7.36$, since $e^{7.36} > 1571$, so $q(u + cu^3) < 0.01\,(7.36 + 0.016 \cdot 398.7) < 0.138$. Smoothness on $(0, X]$ is clear from (3.2), since q and y are smooth functions of x there. □

Lemma 3.5. *$G(x_*) = 0$ and $G'(x_*) = 0$.*

Proof. At $x = x_*$ we have $q = q_*$, $q^2 = 1/5$ and $y = y_* = 5x_*$, so $5f(x_*) = \Phi(5x_*)$. Since $dq/dx = 1 - q$ and $dy/dx = -1/q$,

$$f'(x) = 2q(1-q)\Phi(y) - q\Phi'(y). \tag{3.3}$$

Hence

$$G'(x_*) = 5\Phi'(y_*) - 5f'(x_*) = 5\big[(1+q_*)\Phi'(y_*) - 2q_*(1-q_*)\Phi(y_*)\big],$$

which vanishes by Lemma 3.3(iv) and the definition of d_*. □

3.3. **Proof of Theorem 1.3.** Assume Proposition 3.2. Fix $m \geq 1$ and $Y \in [0, mX]$, and consider the compact set

$$K_Y = \Big\{(x_1, \ldots, x_m) \in [0, X]^m : \sum_i x_i = Y\Big\}.$$

By Lemma 3.4 the function $S(x_1, \ldots, x_m) = \sum_i f(x_i)$ is continuous on K_Y, so it attains its maximum at some point $\xi = (\xi_1, \ldots, \xi_m)$. It suffices to show $S(\xi) \leq \Phi(Y)$. Suppose, to the contrary, that $S(\xi) > \Phi(Y)$. Then $Y > 0$, since $Y = 0$ forces $\xi = 0$ and $S(\xi) = 0 = \Phi(0)$.

Step 1: $5/2 < Y < 31/10$. By (P2) and $f(0) = 0$, $S(\xi) < \frac{1001}{1000}Y$, because some ξ_i is positive. If $Y \notin (5/2, 31/10)$, Lemma 3.3(iii) gives $\Phi(Y) \geq \frac{1001}{1000}Y > S(\xi)$, a contradiction.

Step 2: every positive coordinate of ξ lies in $(0, A)$. Suppose $\xi_j \geq A$. By (P3) and (P2),

$$S(\xi) < \frac{99}{100}\xi_j + \frac{1001}{1000}(Y - \xi_j) = Y\Big(\frac{1001}{1000} - \frac{11}{1000} \cdot \frac{\xi_j}{Y}\Big).$$

By (P1) and Step 1, $\xi_j/Y > (9/10)/(31/10) = 9/31$, and $\frac{1001}{1000} - \frac{11}{1000} \cdot \frac{9}{31} = 1 - \frac{68}{31000} < 1$. Hence $S(\xi) < Y \le \Phi(Y)$ by Lemma 3.3(i), a contradiction.

Step 3: the shape of f' on $(0, A]$. By (P5), f'' is strictly convex on $(0, A]$ and $f''(A) < 0$. Let $N = \{s \in (0, A] : f''(s) < 0\}$, which contains A, and let $a = \inf N \in [0, A)$. If $s \in N$ and $s < s' \le A$, then by convexity $f''(s') \le \max\{f''(s), f''(A)\} < 0$, so $N \supseteq (a, A]$. If $a > 0$ and $0 < s < a$, then $f''(s) > 0$: otherwise, for every $s' \in (s, a)$, strict convexity would give $f''(s') < \max\{f''(s), f''(A)\} \le 0$, contradicting $s' \notin N$. Consequently f' is strictly increasing on $(0, a]$ and strictly decreasing on $[a, A]$ (the first interval being empty if $a = 0$). It follows that if $0 < u < v < A$ and $f'(u) = f'(v)$, then $u < a < v$, and hence $f''(u) > 0 > f''(v)$; indeed $u = a$ is impossible because $f'(v) < f'(a)$ for $v > a$.

Step 4: all positive coordinates of ξ are equal. Let $P = \{i : \xi_i > 0\} \neq \emptyset$. By Step 2 the point $(\xi_i)_{i \in P}$ lies in the open set $(0, A)^P$, and it maximizes the smooth function $\sum_{i \in P} f(x_i)$ on this open set under the constraint $\sum_{i \in P} x_i = Y$, with the remaining coordinates held at 0. By the Lagrange condition, $f'(\xi_i)$ takes the same value for all $i \in P$. By Step 3 the coordinates ξ_i, $i \in P$, take at most two distinct values. Suppose they take two values $u < v$; then $f''(u) > 0 > f''(v)$ by Step 3. If two coordinates $i \neq i'$ in P equal u, the curve $h \mapsto \xi + h(e_i - e_{i'})$ stays in K_Y for small $|h|$, and along it the second derivative of S at $h = 0$ is $2f''(u) > 0$ while the first derivative vanishes; this contradicts the maximality of ξ. So exactly one coordinate i_0 equals u and $k \ge 1$ coordinates equal v. Along the curve that adds h to ξ_{i_0} and subtracts h/k from each coordinate equal to v, the first derivative of S at $h = 0$ is $f'(u) - f'(v) = 0$ and the second derivative is

$$f''(u) + \frac{f''(v)}{k} \;\ge\; f''(u) + f''(v),$$

since $f''(v) < 0$. As f'' is strictly convex on $[u, v]$,

$$\frac{f''(u) + f''(v)}{2} > \frac{1}{v - u} \int_u^v f''(s)\, ds = \frac{f'(v) - f'(u)}{v - u} = 0.$$

Hence the second derivative is positive, again contradicting maximality. Therefore all positive coordinates of ξ are equal.

Step 5: the number of positive coordinates is five. Let $b = |P|$, so that $\xi_i = Y/b$ for $i \in P$ and $S(\xi) = bf(Y/b)$. If $b \le 4$, Step 1 gives $Y/b > 5/8$, and (P4) gives $S(\xi) < Y \le \Phi(Y)$. If $b \ge 6$, then $Y/b < 31/60 < 13/25$, and the same contradiction follows from (P4). Hence $b = 5$, and $x = Y/5 \in (1/2, 31/50)$.

Step 6: the case $b = 5$. By (P1), $x_* \in (1/2, 31/50)$. By (P6), G is strictly convex on $[1/2, 31/50]$, and by Lemma 3.5 it has a stationary point at x_* with $G(x_*) = 0$. Hence $G \ge 0$ on this interval, that is, $5f(x) \le \Phi(5x) = \Phi(Y)$, so $S(\xi) \le \Phi(Y)$. This final contradiction proves Theorem 1.3. □

Remark 3.6. By Lemma 3.5, equality holds in Theorem 1.3 for five entries equal to x_*, and zero entries may be added. For the linear potential $y \mapsto y$ the corresponding inequality $\sum q_i^2 \log(\Lambda/r_i) \le \sum \log(1+r_i)$ for all lists in $(0, \Lambda]$ is equivalent, by taking one entry, to $q^2 \log(\Lambda/r) \le \log(1 + r)$ for $0 < r \le \Lambda$. A numerical computation indicates that this holds precisely for $\Lambda \le 15.6641\ldots$; this is not used in the proofs.

3.4. **Strict contraction below L.** For $0 < \Lambda \le L$ and $0 < r \le \Lambda$ put $x = \log(1 + r)$, $q = r/(1 + r)$ and

$$f_\Lambda(x) = q^2\, \Phi\big(\log(\Lambda/r)\big).$$

Thus $f_L = f$ on $(0, X]$.

Lemma 3.7. *Let $0 < \Lambda < L$. There is $\gamma = \gamma(\Lambda) \in (0,1)$ such that the following holds. For every $m \geq 0$, all $r_1, \dots, r_m \in (0, \Lambda]$ and all $y_1, \dots, y_m$ with $0 \leq y_i \leq \log(\Lambda/r_i)$,*

$$\sum_{i=1}^{m} \Big(\frac{r_i}{1+r_i}\Big)^2 \Phi(y_i) \;\leq\; \gamma\, \Phi\Big(\sum_{i=1}^{m} \log(1+r_i)\Big).$$

Proof. For $0 < r \leq \Lambda$ let $g(r) = f_L(x)/x$ and $h(r) = (f_L(x) - f_\Lambda(x))/x$, where $x = \log(1+r)$. Both are continuous, and $h > 0$ because Φ is strictly increasing and $\log(L/r) > \log(\Lambda/r)$. By Lemma 3.4, $g(r) \to 0$ as $r \to 0$, so there is $r_0 \in (0, \Lambda]$ with $g \leq 1/2$ on $(0, r_0)$. On the compact interval $[r_0, \Lambda]$ let $m_h > 0$ be the minimum of h and M_g the maximum of g; by (P2), $0 < M_g < 2$. Put

$$\theta = 1 - \min\Big\{\frac{1}{2}, \frac{m_h}{M_g}\Big\} \in \Big[\frac{1}{2}, 1\Big).$$

We claim that

$$f_\Lambda(x) \leq \theta f_L(x) + (1-\theta)\frac{x}{2} \qquad (0 < r \leq \Lambda). \tag{3.4}$$

This is equivalent to $h \geq (1-\theta)(g - 1/2)$. On $(0, r_0)$ the right side is at most $0 < h$. On $[r_0, \Lambda]$ the right side is at most $(1-\theta)M_g \leq m_h \leq h$.

Now let $x_i = \log(1+r_i) \in (0, \log(1+\Lambda)] \subset [0, X]$ and $Y = \sum_i x_i$. Since Φ is increasing, $(r_i/(1+r_i))^2 \Phi(y_i) \leq f_\Lambda(x_i)$. Summing (3.4) and using Theorem 1.3 and Lemma 3.3(i),

$$\sum_i f_\Lambda(x_i) \leq \theta\, \Phi(Y) + (1-\theta)\frac{Y}{2} \leq \frac{1+\theta}{2}\, \Phi(Y).$$

The lemma holds with $\gamma = (1+\theta)/2$; for $m = 0$ both sides vanish. □

The pointwise inequality $f_\Lambda < f_L$ alone would not give a factor $\gamma < 1$, since $f_\Lambda(x)/f_L(x) \to 1$ as $x \to 0$. The term $x/2$ in (3.4) compensates for this, because $f_L(x)/x \to 0$.

4. The correlation matrix on forests

Throughout this section $0 < \Lambda < L$, $\gamma = \gamma(\Lambda)$ is as in Lemma 3.7, and F is a finite forest with activities $0 < \lambda_v \leq \Lambda$, with a root chosen in each component. Put

$$C_\Lambda = \Lambda(1+\Lambda)^2 \sup_{y \geq 0} e^{-y}\Phi(y) < \infty, \qquad \eta_\Lambda = \Big(1 + \frac{\sqrt{C_\Lambda}}{1-\sqrt{\gamma}}\Big)^2. \tag{4.1}$$

4.1. **Decay of squared correlations.** For $v \in V$ and $h \geq 0$ let $D_h(v)$ be the set of descendants of v at distance h from v, and put

$$A_h(v) = \sum_{w \in D_h(v)} Q(v,w)^2\, \Phi(y_w), \qquad S_h(v) = \sum_{u \in D_h(v)} R_{vu}^2.$$

Lemma 4.1. *For every $v \in V$ and every $h \geq 0$, $A_h(v) \leq \gamma^h\, \Phi(y_v)$.*

Proof. For every vertex w, apply Lemma 3.7 to the list $(r_u)_{u\in\mathcal{C}(w)}$ with y_u in place of y_i. This is permitted, since $r_u \le \lambda_u \le \Lambda$ and $0 \le y_u = \log(\lambda_u/r_u) \le \log(\Lambda/r_u)$ by Lemma 2.2(i), and since $\sum_{u\in\mathcal{C}(w)} \log(1+r_u) = y_w$. It gives

$$\sum_{u\in\mathcal{C}(w)} q_u^2\, \Phi(y_u) \le \gamma\, \Phi(y_w). \tag{4.2}$$

The case $h = 0$ is trivial. For $h \ge 1$, every $u \in D_h(v)$ is a child of a unique $w \in D_{h-1}(v)$, and $Q(v,u) = Q(v,w)q_u$. By (4.2), $A_h(v) \le \gamma A_{h-1}(v)$, and the claim follows by induction. □

Proposition 4.2. *For every $v \in V$ and every $h \ge 1$, $S_h(v) \le C_\Lambda\, \gamma^{h-1}$.*

Proof. By (2.2), $S_h(v) = s_v \sum_{u\in D_h(v)} Q(v,u)^2/s_u$. Write each $u \in D_h(v)$ as a child of $w \in D_{h-1}(v)$. By Lemma 2.2(iii), $Q(v,u)^2/s_u \le (1+\Lambda)^2 Q(v,w)^2 q_u$. The elementary inequality $r/(1+r) \le \log(1+r)$ gives $\sum_{u\in\mathcal{C}(w)} q_u \le y_w \le \Phi(y_w)$. Therefore, by Lemma 4.1,

$$S_h(v) \le (1+\Lambda)^2 s_v\, A_{h-1}(v) \le (1+\Lambda)^2 \gamma^{h-1} s_v \Phi(y_v).$$

Finally $s_v\Phi(y_v) \le \Lambda e^{-y_v}\Phi(y_v)$ by Lemma 2.2(iii), and the supremum in (4.1) is finite because Φ is a polynomial. □

The last generation is treated separately in the proof because the variance s_u in the denominator absorbs one of the two factors q_u.

4.2. **A triangular factorization.** The following factorization holds for every forest with positive activities; it does not use $\Lambda < L$. The inverse correlation matrix of a tree model has the corresponding sum-of-squares form, which is the starting point of the approximate inverse method of [4, Proposition 29, Lemma 41].

Order V so that every non-root vertex comes after its parent. Let B be the $V \times V$ matrix with $B_{u,u^-} = R_{u,u^-}$ for every non-root u and all other entries 0. It is strictly lower triangular, hence nilpotent. Let $\mathcal{D}$ be the diagonal matrix with $\mathcal{D}_{vv} = 1$ for roots and $\mathcal{D}_{uu} = 1 - R^2_{u,u^-}$ otherwise.

Lemma 4.3. (i) *For $h \ge 1$, $(B^h)_{uw} = R_{uw}$ if w is the ancestor of u at distance h, and $(B^h)_{uw} = 0$ otherwise.*

(ii) *With $K = (I-B)^{-1} = \sum_{h\ge 0} B^h$ one has $R = K\mathcal{D}K^{\mathsf{T}}$, and $0 < \mathcal{D}_{uu} \le 1$ for every u.*

Proof. (i) For $h = 1$ this is the definition. If it holds for $h-1$, then

$$(B^h)_{uw} = \sum_k B_{uk}(B^{h-1})_{kw} = R_{u,u^-}(B^{h-1})_{u^-,w},$$

which vanishes unless w is the ancestor of u^- at distance $h-1$. In that case (2.2) (and the symmetry of R) gives

$$\begin{aligned} R_{u,u^-}R_{u^-,w} &= \Big(-\sqrt{\tfrac{s_{u^-}}{s_u}}\, q_u\Big)\Big((-1)^{h-1}\sqrt{\tfrac{s_w}{s_{u^-}}}\, Q(w,u^-)\Big) \\ &= (-1)^h \sqrt{\tfrac{s_w}{s_u}}\, Q(w,u) = R_{uw}. \end{aligned}$$

(ii) Let $Y_v = (X_v - a_v)/\sqrt{s_v}$, and define $\varepsilon_v = Y_v$ for roots and $\varepsilon_u = Y_u - R_{u,u^-}Y_{u^-}$ otherwise, so that $\varepsilon = (I-B)Y$. Since X_{u^-} takes two values, $\mathbb{E}[Y_u \mid X_{u^-}]$ is an affine function of Y_{u^-}, and comparing means and covariances gives $\mathbb{E}[Y_u \mid X_{u^-}] =$

$R_{u,u^-}Y_{u^-}$. Let W_u be the set of vertices preceding u in the chosen order. No vertex of T_u precedes u, so $W_u \cap T_u = \emptyset$, and $u^- \in W_u$. For a non-root u, Lemma 2.1, applied at $w = u^-$ to the component T_u of $F - u^-$, gives $\mathbb{E}[Y_u \mid X_{W_u}] = \mathbb{E}[Y_u \mid X_{u^-}]$, hence $\mathbb{E}[\varepsilon_u \mid X_{W_u}] = 0$. For a root v, all vertices of its component come after v, so X_{W_v} is independent of Y_v and $\mathbb{E}[\varepsilon_v \mid X_{W_v}] = \mathbb{E}[Y_v] = 0$. Every $\varepsilon_{u'}$ with u' preceding u is a function of X_{W_u}, so the ε_u are pairwise uncorrelated. Moreover $\mathbb{E}\varepsilon_u^2 = 1 - 2R_{u,u^-}^2 + R_{u,u^-}^2 = 1 - R_{u,u^-}^2$ for non-roots. Thus $\mathrm{Cov}(\varepsilon) = \mathcal{D}$ and $R = \mathrm{Cov}(Y) = K\mathcal{D}K^{\mathsf{T}}$. Finally $|R_{u,u^-}| < 1$, since X_u is not almost surely an affine function of X_{u^-}: conditionally on $X_{u^-} = 0$, both values of X_u have positive probability. □

4.3. **Proof of Theorem 1.2.** By Lemma 4.3(i), every row of B^h has at most one nonzero entry. Hence $(B^h)^{\mathsf{T}}B^h$ is diagonal, with entry $S_h(w)$ at w, and Proposition 4.2 gives $\|B^h\|_2^2 = \max_w S_h(w) \le C_\Lambda \gamma^{h-1}$ for $h \ge 1$. Therefore

$$\|K\|_2 \le \sum_{h\ge 0} \|B^h\|_2 \le 1 + \sqrt{C_\Lambda} \sum_{h\ge 1} \gamma^{(h-1)/2} = 1 + \frac{\sqrt{C_\Lambda}}{1-\sqrt{\gamma}},$$

and by Lemma 4.3(ii), $\|R\|_2 \le \|K\|_2^2 \|\mathcal{D}\|_2 \le \eta_\Lambda$. □

4.4. **Complete spectral independence.**

Lemma 4.4. *Let F be a finite forest with positive activities and let Ψ be the influence matrix of the unpinned measure $\mu_{F,\lambda}$. Then $\Psi = S^{-1/2}(|R| - I)S^{1/2}$ with $S = \mathrm{diag}(s_v)$, and $\rho(\Psi) = \lambda_{\max}(R) - 1$.*

Proof. All coordinates are nonconstant, so for $u \ne v$

$$\Psi(u,v) = \frac{|\mathrm{Cov}(X_u, X_v)|}{s_u} = s_u^{-1/2}\, |R_{uv}|\, s_v^{1/2},$$

which is the asserted similarity. The matrix $|R| - I$ is symmetric with nonnegative entries, so its spectral radius is its largest eigenvalue, which equals $\lambda_{\max}(R) - 1$ by Lemma 2.4. □

Proof of the spectral independence in Theorem 1.1(a). Let $\phi \in (0,1]^V$ and let σ be a feasible pinning on $\Lambda' \subseteq V$ for $\nu = \mu_{F,\lambda}^{(\phi)}$. Let U be as in Lemma 2.6. By Definition 2.5, the rows and columns of Ψ_{ν^σ} indexed by the almost surely constant coordinates $V \setminus (\Lambda' \cup U)$ vanish, and by Lemma 2.6 the remaining principal submatrix is the influence matrix of the hard-core measure on $F[U]$ with activities $\phi_v \lambda_v \le \Lambda$. If $U = \emptyset$, then $\Psi_{\nu^\sigma} = 0$. Otherwise Lemma 4.4 and Theorem 1.2, applied to $F[U]$, give $\rho(\Psi_{\nu^\sigma}) \le \eta_\Lambda - 1 < \eta_\Lambda$. □

5. The spectral gap

We use two published results. The first is the boosting theorem of Chen, Feng, Yin and Zhang. For $\theta \in (0,1)$ write $\nu^{(\theta)}$ for $\nu^{(\phi)}$ with $\phi \equiv \theta$, and let $\mathrm{gap}_{\min}(\nu)$ be the minimum of $\mathrm{gap}(\nu^\sigma)$ over all $\Lambda' \subseteq V$ and all feasible $\sigma \in \{0,1\}^{\Lambda'}$, with the conventions of Section 2.

Theorem 5.1 ([3, Theorem 1.9])**.** *If ν is completely η-spectrally independent, then for every $\theta \in (0,1)$,*

$$\mathrm{gap}(\nu) \;\ge\; \Big(\frac{\theta}{2}\Big)^{2\eta+7} \mathrm{gap}_{\min}\big(\nu^{(\theta)}\big).$$

The second is the variance tensorization of Efthymiou, Hayes, Štefankovič and Vigoda for small activities. In [6, Section 3] it is proved, for every finite tree with arbitrary vertex activities in $(0, 1.1]$ [6, (11)], that

$$\mathrm{Var}_\mu(g) \le \sum_{v \in V} \kappa(\lambda_v)\, \mathbb{E}_\mu\big[\mathrm{Var}_\mu(g \mid X_{V\setminus\{v\}})\big], \qquad \kappa(s) = 1000(1+s)^2 e^{1.3s}, \tag{5.1}$$

for every real function g on the independent sets; see [6, (12) and Lemma 3.1]. (In [6, Definition 2.1] functions are nonnegative; since both sides of (5.1) are invariant under adding constants, this is no restriction.)

Lemma 5.2. *Inequality* (5.1) *holds for every finite forest with activities in* $(0, 1.1]$.

Proof. The hard-core measure on a forest with components $T_1, \dots, T_k$ is the product of the hard-core measures on the T_j. For a product measure one has $\mathrm{Var}(g) \le \sum_j \mathbb{E}[\mathrm{Var}_j(g)]$, where Var_j denotes the variance in the coordinates of T_j given the others (the Efron–Stein inequality). For fixed values outside T_j, apply (5.1) on T_j to the resulting function of X_{T_j}; by independence, its single-site conditional variances are those of g under μ. Taking expectations and summing over j gives the claim. □

Proof of the spectral gap bound in Theorem 1.1(a). Let $\theta = 1/(2\max\{1, \Lambda\})$ and $\nu = \mu_{F,\lambda}$. By Section 4, ν is completely η_Λ-spectrally independent. Let σ be a feasible pinning for $\nu^{(\theta)}$ on $\Lambda' \subseteq V$, and let U be as in Lemma 2.6. If $U = \emptyset$, then $|\Omega(\nu^{(\theta)\sigma})| = 1$ and the gap is 1 by convention. Otherwise, by Lemma 2.6, $\nu^{(\theta)\sigma}$ is the hard-core measure on $F[U]$ with activities $\theta\lambda_v \le 1/2$, extended by constants. For a function g on its support, Lemma 5.2 gives

$$\mathrm{Var}(g) \le \kappa(\tfrac12) \sum_{v \in U} \mathbb{E}\big[\mathrm{Var}(g \mid X_{V\setminus\{v\}})\big] = n\,\kappa(\tfrac12)\,\mathcal{E}(g,g),$$

because κ is increasing and the conditional variances at the coordinates outside U vanish. By (2.3), $\mathrm{gap}(\nu^{(\theta)\sigma}) \ge 1/(n\kappa(1/2))$. Theorem 5.1 now gives

$$\mathrm{gap}(\nu) \ge \frac{c_\Lambda}{n}, \qquad c_\Lambda = \frac{(\theta/2)^{2\eta_\Lambda+7}}{\kappa(1/2)}. \qquad \square$$

Remark 5.3. Fix $\lambda \in (0, L)$ and $\Delta \ge 3$, and consider forests of maximum degree at most Δ with activity λ at every vertex. Under every feasible pinning, each value of an unpinned coordinate that has positive conditional probability has conditional probability at least $\lambda(1+\lambda)^{-\Delta-1}$, so these measures are marginally bounded in the sense of [5], uniformly in the forest. In this setting Theorem 1.1(a) can be combined with [5, Theorem 1.12] to obtain mixing time bounds of order $n \log n$. When the activities are not bounded away from zero, as is allowed in Theorem 1.1(a), no such lower bound on the marginals holds. We do not pursue this here.

6. The upper threshold

In this section $\lambda \ge L$ is a constant activity used at every vertex. Since $s \mapsto s(1-s)^{-6}$ increases from 0 to ∞ on $(0, 1)$, there is a unique $p \in (0, 1)$ with

$$p = \lambda(1-p)^6, \tag{6.1}$$

and $p \ge q_*$, that is, $5p^2 \ge 1$, because $\lambda \ge L = q_*(1-q_*)^{-6}$. Put $\omega = p/(1-p)$. Then $1 + \omega = (1-p)^{-1}$ and

$$\lambda(1+\omega)^{-5} = \omega, \qquad \lambda(1+\omega)^{-6} = p. \tag{6.2}$$

Lemma 6.1 ([4, Lemma 57]; see [9, Section 5])**.** *Let $\lambda > 0$ and $d \geq 3$. For every $\delta \in (0,1)$ there is a finite rooted tree T_δ such that, for the hard-core measure on T_δ with activity λ at every vertex, the occupation probability of the root differs by less than δ from the unique solution in $(0,1)$ of $s = \lambda(1-s)^d$.*

We apply Lemma 6.1 with $d = 6$, so that the target value is p. Let ω_δ be the occupation odds of the root of T_δ; then $\omega_\delta \to \omega$ as $\delta \to 0$.

6.1. **The construction.** For $H \geq 1$ let $\mathcal{C}_H$ be the rooted tree of height H in which the root has six children and every other vertex at depth less than H has five children. Its leaves are the vertices at depth H. Let $T_{H,\delta}$ be the tree obtained from $\mathcal{C}_H$ by identifying every leaf of $\mathcal{C}_H$ with the root of a separate copy of T_δ. This is a finite tree, and we consider the hard-core measure on it with activity λ at every vertex.

Lemma 6.2. *The law of the restriction of X to the vertices of $\mathcal{C}_H$ is the hard-core measure on $\mathcal{C}_H$ with activity λ at the internal vertices and activity ω_δ at the leaves.*

Proof. For a leaf ℓ of $\mathcal{C}_H$ let $T^{(\ell)}$ be the attached copy of T_δ, rooted at ℓ, and let Z^1 and Z^0 be the total weights of the independent sets of T_δ that contain, respectively do not contain, its root; thus $Z^1/Z^0 = \omega_\delta$. An independent set of $T_{H,\delta}$ is the same as an independent set J of $\mathcal{C}_H$ together with, for every leaf ℓ, an independent set of $T^{(\ell)}$ whose intersection with $\{\ell\}$ agrees with J. Its weight is $\lambda^{|J'|}$ times the product of the weights of the sets in the copies, where J' is the set of internal vertices of $\mathcal{C}_H$ in J. Summing over the sets in the copies, the total weight of the configurations that restrict to J on $\mathcal{C}_H$ is $\lambda^{|J'|}(Z^1)^{|J\setminus J'|}(Z^0)^{k-|J\setminus J'|}$, where k is the number of leaves of $\mathcal{C}_H$. Dividing by $(Z^0)^k$ gives the weight $\lambda^{|J'|}\omega_\delta^{|J\setminus J'|}$. □

Let $\nu_{H,\omega'}$ denote the hard-core measure on $\mathcal{C}_H$ with activity λ at internal vertices and $\omega' > 0$ at the leaves, and $R^{(H,\omega')}$ its correlation matrix. The means, variances and covariances of the occupation indicators under $\nu_{H,\omega'}$ are rational functions of ω' with denominators that are positive for $\omega' > 0$, and all variances are positive. Hence the entries of $R^{(H,\omega')}$ are continuous in $\omega' > 0$.

Lemma 6.3. *In $\nu_{H,\omega}$ every vertex has occupation probability $p/(1+p)$, and $R^{(H,\omega)}_{u,u^-} = -p$ for every non-root vertex u.*

Proof. Root $\mathcal{C}_H$ at its root. The leaves have odds ω, and by (6.2) and Lemma 2.2(i) every non-root vertex has odds $\lambda(1+\omega)^{-5} = \omega$, by induction on the height. Hence $q_u = \omega/(1+\omega) = p$ for every non-root u, and the root has odds $\lambda(1+\omega)^{-6} = p$, so its occupation probability is $p/(1+p)$. By Lemma 2.2(ii), $a_u = (1-a_{u^-})p$ for every non-root u, and by induction on the depth $a_u = (1-p/(1+p))p = p/(1+p)$. All variances are therefore equal, and (2.2) gives $R_{u,u^-} = -q_u = -p$. □

Lemma 6.4. $\lambda_{\max}\big(R^{(H,\omega)}\big) \geq 1 + 6p^2\sum_{j=0}^{H-1}(5p^2)^j \geq 1 + \frac{6H}{5}$.

Proof. Write $R = R^{(H,\omega)}$. By Lemma 4.3 and Lemma 6.3, $R = K\mathcal{D}K^{\mathsf{T}}$, where $K = (I-B)^{-1}$ is unit lower triangular, $B_{u,u^-} = -p$, and $\mathcal{D}$ is diagonal with entry 1 at the root and $1-p^2 > 0$ elsewhere. Hence R is invertible and

$$\xi^{\mathsf{T}}R^{-1}\xi = \big\|\mathcal{D}^{-1/2}(I-B)\xi\big\|^2 = \xi_{\mathrm{root}}^2 + \sum_{u\neq\mathrm{root}} \frac{(\xi_u + p\,\xi_{u^-})^2}{1-p^2} \qquad (\xi \in \mathbb{R}^{\mathcal{C}_H}).$$

Take $\xi_u = (-p)^{\mathrm{depth}(u)}$, the vector of [4, (21)] in the present setting. Then $\xi_u + p\,\xi_{u^-} = 0$ for every non-root u, so $\xi^{\mathsf{T}} R^{-1}\xi = 1$. By the Rayleigh principle, $\lambda_{\max}(R) = \lambda_{\min}(R^{-1})^{-1} \geq \|\xi\|^2$. There are $6\cdot 5^{j-1}$ vertices at depth $j \geq 1$, so

$$\|\xi\|^2 = 1 + \sum_{j=1}^{H} 6\cdot 5^{j-1}p^{2j} = 1 + 6p^2 \sum_{j=0}^{H-1} (5p^2)^j \geq 1 + 6p^2 H \geq 1 + \frac{6H}{5},$$

using $5p^2 \geq 1$. □

6.2. **Proof of Theorem 1.1(b).** We first record a general upper bound for the spectral gap.

Lemma 6.5. *Let ν be a probability measure on $\{0,1\}^V$, $|V| = n$, without almost surely constant coordinates, and let R be its correlation matrix. Then* $\mathrm{gap}(\nu) \leq 1/(n\,\lambda_{\max}(R))$.

Proof. Let ξ be a unit eigenvector of R for $\lambda_{\max}(R)$ and put $g = \sum_v \xi_v (X_v - a_v)/\sqrt{s_v}$. Then $\mathrm{Var}(g) = \xi^{\mathsf{T}} R \xi = \lambda_{\max}(R)$. Given $X_{V\setminus\{v\}}$, the function g differs from $\xi_v X_v/\sqrt{s_v}$ by a constant, so by the law of total variance

$$\mathcal{E}(g,g) = \frac{1}{n}\sum_v \frac{\xi_v^2}{s_v}\,\mathbb{E}\big[\mathrm{Var}(X_v \mid X_{V\setminus\{v\}})\big] \leq \frac{1}{n}\sum_v \xi_v^2 = \frac{1}{n}.$$

The claim follows from (2.3). □

Proof of Theorem 1.1(b). Let $M > 0$. Choose $H \geq 1$ with $6H/5 > M$, so that $\lambda_{\max}(R^{(H,\omega)}) > M+1$ by Lemma 6.4. Since $R^{(H,\omega')}$ depends continuously on ω', the largest eigenvalue of a symmetric matrix depends continuously on its entries, and $\omega_\delta \to \omega$ as $\delta \to 0$, there is $\delta \in (0,1)$ with $\lambda_{\max}(R^{(H,\omega_\delta)}) > M+1$. Let $T = T_{H,\delta}$, let μ be its hard-core measure with activity λ, and let R_T be its correlation matrix. By Lemma 6.2, the principal submatrix of R_T indexed by the vertices of $\mathcal{C}_H$ is $R^{(H,\omega_\delta)}$, so $\lambda_{\max}(R_T) \geq \lambda_{\max}(R^{(H,\omega_\delta)}) > M+1$. By Lemma 4.4, $\rho(\Psi_\mu) = \lambda_{\max}(R_T) - 1 > M$, and by Lemma 6.5, $n\cdot\mathrm{gap}(\mu) \leq 1/\lambda_{\max}(R_T) < 1/M$. □

Remark 6.6. For $\lambda = L$ the bound of Lemma 6.4 grows linearly in H, while for $\lambda > L$ it grows exponentially. In both cases the correlation matrices of unbounded norm are those of the free-boundary measures on the constructed trees, so Lemma 6.5 applies directly and no comparison between spectral independence and the spectral gap is needed. With $\mathcal{C}_H$ replaced by a core of branching number $b \geq 2$ (root with $b+1$ children) and $d = b+1$ in Lemma 6.1, the same argument applies whenever $bp^2 \geq 1$ with $p = \lambda(1-p)^{b+1}$; for $b = 2$ this is the setting of [4, Section 5.2].

7. Verification of the scalar estimates

In this section we prove Proposition 3.2. The constants in (P1) and all statements on intervals bounded away from $x = 0$ are verified by interval arithmetic with rational endpoints; the behaviour near $x = 0$ is handled analytically.

7.1. **Formulas in the variable q.** By (3.2), the map $x \mapsto q = 1 - e^{-x}$ is an increasing bijection from $(0, X]$ onto $(0, L/(1+L)]$, with $x = -\log(1-q)$. For a function of (q, y) the derivative with respect to x is obtained by applying the operator $(1-q)\partial_q - q^{-1}\partial_y$. Thus

$$\frac{f(x)}{x} = \frac{q^2\,\Phi(y)}{-\log(1-q)}, \tag{7.1}$$

$$f''(x) = 2(1-q)(1-2q)\,\Phi(y) - 3(1-q)\,\Phi'(y) + \Phi''(y), \tag{7.2}$$

$$q^2 f''''(x) = a_0 + a_1 y + a_2 y^2 + a_3 y^3, \tag{7.3}$$

$$G''(x) = 50c\,(15x - 2z) - 5f''(x), \tag{7.4}$$

where $\Phi''(y) = 2c(3y - 2z)$ and

$$a_0 = (q-1)\big(15cq^2z^2 - 6cqz^2 - 28cqz - 2cz^2 - 4cz + 12c + 15q^2 - 6q - 2\big),$$
$$a_1 = 2(q-1)\big(8cq^3z^2 - 7cq^2z^2 - 30cq^2z + 12cqz + 21cq + 4cz + 3c + 8q^3 - 7q^2\big),$$
$$a_2 = -c\,(q-1)\big(32q^3z - 28q^2z - 45q^2 + 18q + 6\big),$$
$$a_3 = 2c\,q^2(q-1)(8q-7).$$

Formula (7.2) follows from (3.3) by one more application of the operator, and (7.3) by three more; the right side of (7.3) is a polynomial in q and y because the negative powers of q cancel. Formula (7.3) has also been confirmed by exact symbolic computation, as part of the second program described below. In (7.4) we used $\Phi''(5x) = 2c(15x - 2z)$. The points $x = 13/25, 5/8, 1/2, 31/50$ are bracketed in the variable q by

$$\begin{aligned} -\log\tfrac{594}{1000} > \tfrac{13}{25}, \qquad -\log\tfrac{5353}{10000} < \tfrac{5}{8}, \\ -\log\tfrac{607}{1000} < \tfrac{1}{2}, \qquad -\log\tfrac{537}{1000} > \tfrac{31}{50}, \end{aligned} \tag{7.5}$$

that is, $x(406/1000) > 13/25$, $x(4647/10000) < 5/8$, $x(393/1000) < 1/2$ and $x(463/1000) > 31/50$.

7.2. **Interval arithmetic.** Two independently implemented programs, written in Python with the standard library only, carry out the verification. Every interval endpoint is an integer multiple of a fixed unit, 10^{-40} in the first program and 2^{-170} in the second. Sums are exact; products, quotients, integer powers and square roots (via integer square roots) are rounded outward. No floating-point number enters any decision. Logarithms of positive intervals are enclosed as follows. In the first program the argument is scaled by a power of two into $[1, 2)$, and $\log u = 2\sum_{j\ge 0} w^{2j+1}/(2j+1)$ with $w = (u-1)/(u+1) \in [0, 1/3)$ is truncated after 55 terms; the remainder lies in $[0,\, 2w^{111}/(111(1-w^2))]$ and is added to the upper endpoint. In the second program the argument is scaled into $(1/2, 1]$, and $-\log(1-w) = \sum_{k\ge 1} w^k/k$ with $w \in [0, 1/2)$ is truncated after 180 terms, with remainder in $[0,\, w^{181}/(181(1-w))]$. In both programs $\log 2$ is enclosed by the same series, and the scaling contributes an integer multiple of it. The constants q_*, L, x_*, y_*, d_*, D, t, z are computed from (3.1) in this arithmetic, so the intervals used below contain the exact constants.

An inequality $P(q) > 0$ on a closed interval $[q_0, q_1]$ is verified by evaluating an enclosure of P over the whole interval, with y and x enclosed through (3.2). If the lower endpoint of the enclosure is positive, the interval is accepted; otherwise it is split at an interior point into two closed subintervals sharing that point, and the

procedure is repeated. Failure to accept before a fixed maximum depth stops the program with an error. The accepted intervals cover $[q_0, q_1]$; the second program checks this explicitly by sorting the accepted intervals and comparing consecutive endpoints exactly.

7.3. **Proof of Proposition 3.2.** *(P1).* Both programs compute enclosures of the constants and check, by comparing endpoints, that $D > 0$, $D^2 < c$, $1/2 < x_* < 31/50$, $L/(1+L) < 941/1000$ and $\log(5/2) > 9/10$. The enclosure of z obtained by each program has rational endpoints in the interval $(2.8310, 2.8311)$; these endpoints are part of the output (Appendix A). Finally $A < X$ because $L > 3/2$.

(P2)–(P6) away from $x = 0$. Table 7.1 lists the verified inequalities, written in terms of q by (7.1)–(7.4), the q-intervals, and the number of interval evaluations performed by each program. The upper endpoint $941/1000$ covers $x = X$ by (P1), the endpoint $3/5$ corresponds to $x = A$, and (7.5) shows that the q-intervals for (P4) and (P6) cover the required x-intervals. In addition, an enclosure of (7.2) at the single point $q = 3/5$ shows $f''(A) < 0$.

TABLE 7.1. Interval verifications. The last two columns give the number of interval evaluations, including those of intervals that were split. The largest bisection depth reached is 15.

Claim	Inequality	q-interval	Program 1	Program 2
(P2)	$\frac{1001}{1000} - f(x)/x > 0$	$[\frac{1}{100}, \frac{941}{1000}]$	851	851
(P3)	$\frac{99}{100} - f(x)/x > 0$	$[\frac{3}{5}, \frac{941}{1000}]$	17	17
(P4)	$1 - f(x)/x > 0$	$[\frac{1}{100}, \frac{406}{1000}]$	91	91
(P4)	$1 - f(x)/x > 0$	$[\frac{4647}{10000}, \frac{941}{1000}]$	465	465
(P5)	$q^2 f''''(x) > 0$	$[\frac{1}{20}, \frac{3}{5}]$	19	73
(P6)	$G''(x) > 0$	$[\frac{393}{1000}, \frac{463}{1000}]$	1	7

(P2) and (P4) near $x = 0$. For $0 < q \le 1/100$, Lemma 3.4 gives $f(x)/x < 0.138$, which proves both on the corresponding range $0 < x \le -\log(99/100)$.

(P5) near $x = 0$. Let $0 < q \le 1/20$. Then $y \ge \log(19L) > 0$. The coefficient $a_3 = 2cq^2(1-q)(7-8q)$ is nonnegative. Enclosing a_0, a_1, a_2 over the whole interval $q \in [0, 1/20]$ gives

$$a_0 > 2.0931, \qquad a_2 > 0.0864, \qquad |a_1| < 0.5486.$$

Hence for every $y \ge 0$,

$$q^2 f''''(x) \ge 2.0931 - 0.5486\, y + 0.0864\, y^2 > 0,$$

since the discriminant $0.5486^2 - 4 \cdot 2.0931 \cdot 0.0864 < 0.31 - 0.72 < 0$. Thus $f''''(x) > 0$ for $0 < q \le 1/20$, which together with Table 7.1 proves (P5).

This completes the proof of Proposition 3.2, and hence of Theorem 1.3. □

Remark 7.1. In the first program the smallest certified lower bounds of the left sides in Table 7.1 are about $1.07 \cdot 10^{-5}$ for (P2), 0.023 for (P3), $1.2 \cdot 10^{-4}$ and $1.3 \cdot 10^{-6}$ for the two ranges of (P4), 0.032 for (P5) and 0.55 for (P6). Both programs accept every inequality of Table 7.1.

Appendix A. Data availability

The two programs of Section 7 and their output certificates are provided as ancillary files accompanying this paper, together with a README file that describes how to run them. The first program is `check_cubic_intervals.py`, and the second is `check_independent.py`. The first program prints a certificate listing the enclosures of q_*, L and z, the bound of Lemma 3.4 at $q = 1/100$, the coefficient bounds used for (P5), and, for each line of Table 7.1, the number of evaluations, the maximal depth and the smallest certified margin, all as exact rational numbers. The second program, which does not import the first, records the enclosures of the same constants and the same kind of data for each line of Table 7.1; in addition, it generates the polynomial in (7.3) by symbolic differentiation, compares it coefficient by coefficient with the displayed formulas, and records every accepted subinterval with its certified margin.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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